

Orthogonal tensors

An orthogonal tensor $\underline{\underline{Q}} \in \mathcal{V}^2$ is a linear transformation satisfying

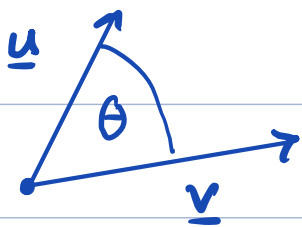
$$\underline{\underline{Q}} \underline{u} \cdot \underline{\underline{Q}} \underline{v} = \underline{u} \cdot \underline{v}$$

for all $\underline{u}, \underline{v} \in \mathcal{V}$

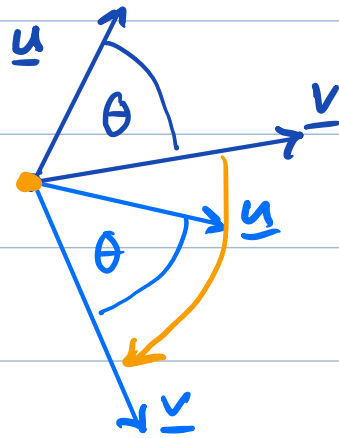
$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

$\Rightarrow$ preserves length & angle

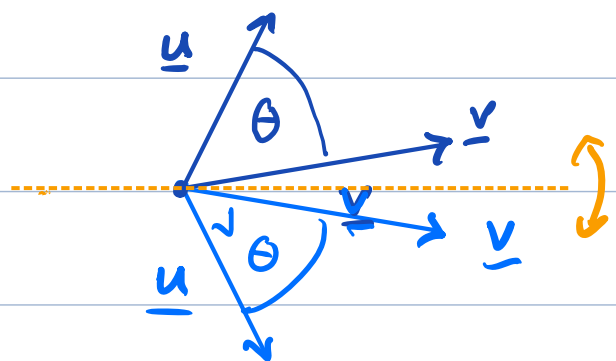
only two possible operations:



rotation



reflection



Properties of orthogonal matrices:

$$\begin{aligned}\underline{\underline{Q}}^T &= \underline{\underline{Q}}^{-1} \\ \underline{\underline{Q}}^T \underline{\underline{Q}} &= \underline{\underline{Q}} \underline{\underline{Q}}^T = \underline{\underline{I}} \\ \det(\underline{\underline{Q}}) &= \pm 1\end{aligned}$$

Show last: $1 = \det(\underline{\underline{I}}) = \det(\underline{\underline{Q}}^T \underline{\underline{Q}})$

$$= \det(\underline{\underline{Q}}^T) \det(\underline{\underline{Q}}) = \det(\underline{\underline{Q}})^2$$
$$\Rightarrow \det(\underline{\underline{Q}}) = \pm 1$$

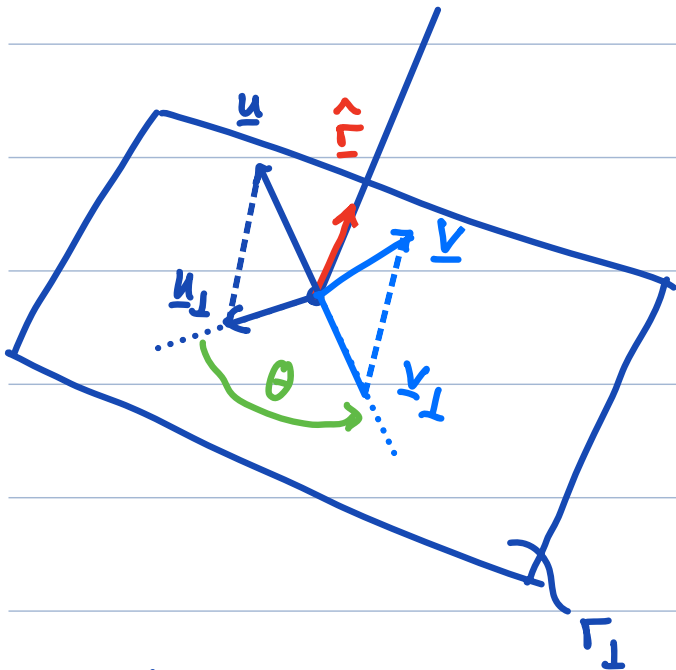
If $\det(\underline{\underline{Q}}) = 1 \Rightarrow$ rotation

$\det(\underline{\underline{Q}}) = -1 \Rightarrow$ reflection

In mechanics we are mostly concerned with rotations.

Rotation Matrices

$$\underline{v} = Q(\hat{r}, \theta) \underline{u}$$



$\hat{r}$ = axis of rotation

$r_{\perp}$ = plane $\perp$ to $\hat{r}$

θ = counter clock wise angle

$$\underline{u} = \underline{u}_{\parallel} + \underline{u}_{\perp}$$

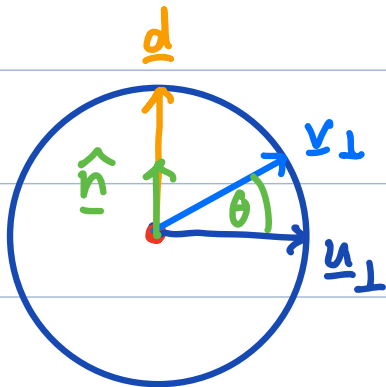
$$\underline{v} = \underline{v}_{\parallel} + \underline{v}_{\perp}$$

$$\underline{v}_{\parallel} = \underline{u}_{\parallel} = (\underline{u} \cdot \hat{r}) \hat{r} = (\underbrace{\hat{r} \otimes \hat{r}}_{P_{\parallel}}) \underline{u}$$

$$\underline{u}_{\perp} = (\underbrace{\mathbb{I} - \hat{r} \otimes \hat{r}}_{P_{\perp}}) \underline{u}$$

What is $\underline{v}_{\perp}$?

looking onto $r_{\perp}$ $\underline{d} = \hat{r} \times \underline{u} = \hat{r} \times (\underline{u}_{\parallel} + \underline{u}_{\perp})$



$$\begin{aligned} &= \hat{r} \times \underline{u}_{\parallel} + \hat{r} \times \underline{u}_{\perp} \\ &= |\hat{r}| |\underline{u}_{\perp}| \sin \phi \hat{n} \end{aligned} \quad \begin{aligned} &\hat{r} \perp \underline{u}_{\perp} \\ &\phi = 90^\circ \end{aligned}$$

$$= |\underline{u}_{\perp}| \hat{n} \Rightarrow |\underline{d}| = |\underline{u}_{\perp}|$$

$\underline{d} \perp \underline{u}_{\perp} \Rightarrow$ basis in $r_{\perp}$

$$\Rightarrow \underline{v}_{\perp} = \cos \theta \underline{u}_{\perp} + \sin \theta \underline{d}$$

Rotated vector:

$$\underline{v} = \underline{v}_{\parallel} + \underline{v}_{\perp} = (\hat{r} \otimes \hat{r}) \underline{u} + \cos \theta (\mathbb{I} - \hat{r} \otimes \hat{r}) \underline{u} + \sin \theta \hat{r} \times \underline{u}$$

Can we write: $\underline{v} = \underline{Q}(\hat{r}, \theta) \underline{u}$?

Axial Tensors

Need to write $\underline{\Gamma} \times \underline{u} = \underline{\underline{R}} \underline{u}$!

$$\underline{\underline{R}} \underline{u} = R_{ij} u_j \underline{e}_i \quad \text{and} \quad \underline{\Gamma} \times \underline{u} = \epsilon_{mnl} \Gamma_m u_n \underline{e}_l$$

$$R_{ij} u_j \underline{e}_i = \epsilon_{mnl} \Gamma_m u_n \underline{e}_l$$

all indices are dummy $\Rightarrow$ rename

$$l \rightarrow i: R_{ij} u_j \underline{e}_i = \epsilon_{mni} \Gamma_m u_n \underline{e}_i$$

$$R_{\color{red}{i}j} u_j = \epsilon_{mni} \color{red}{\Gamma}_m u_n \quad \color{red}{i} = \text{free index}$$

$$n \rightarrow j: R_{ij} u_j = \epsilon_{mji} \Gamma_m u_j$$

$$R_{ij} = \epsilon_{mji} \Gamma_m \quad i, j = \text{free} \quad m = \text{dummy}$$

$$m \rightarrow k: R_{ij} = \epsilon_{kji} \Gamma_k$$

$$\text{prop. of } \epsilon: \epsilon_{kji} = -\epsilon_{jki} = \epsilon_{ikj}$$

$$\boxed{R_{ij} = \epsilon_{ikj} \Gamma_k}$$

$$\text{tr}(\underline{\underline{R}}) = 0$$

$$\underline{\underline{R}} = -\underline{\underline{R}}^T \quad \text{skew sym.}$$

$$\underline{\underline{R}} = R_{ij} \underline{e}_i \otimes \underline{e}_j = \begin{bmatrix} 0 & -\Gamma_3 & \Gamma_2 \\ \Gamma_3 & 0 & -\Gamma_1 \\ -\Gamma_2 & \Gamma_1 & 0 \end{bmatrix}$$

$$R_{12} = \epsilon_{132} \Gamma_3 = -\Gamma_3 \quad R_{13} = \epsilon_{123} \Gamma_2 = \Gamma_2 \quad R_{23} = \epsilon_{213} = -\Gamma_1$$

Back to rotation

$$\begin{aligned}\underline{v} &= (\underline{r} \otimes \underline{r}) \underline{u} + \cos \theta (\underline{I} - \underline{r} \otimes \underline{r}) \underline{u} + \sin \theta \underline{R} \underline{u} \\ &= \underbrace{[\underline{r} \otimes \underline{r} + \cos \theta (\underline{I} - \underline{r} \otimes \underline{r}) + \sin \theta \underline{R}]}_{\underline{Q}(\underline{r}, \theta)} \underline{u}\end{aligned}$$

Euler-Rodrigues' finite rotation tensor

$$\begin{aligned}\underline{Q}(\underline{r}, \theta) &= \underline{r} \otimes \underline{r} + \cos \theta (\underline{I} - \underline{r} \otimes \underline{r}) + \sin \theta \underline{R} \\ Q_{ij}(\underline{r}, \theta) &= r_i r_j + \cos \theta (\delta_{ij} - r_i r_j) + \sin \theta \epsilon_{ikj} r_k\end{aligned}$$

Example: Rotation tensor around $\underline{e}_3$

$$\underline{Q}(\underline{e}_3, \theta) = \underline{e}_3 \otimes \underline{e}_3 + \cos \theta (\underline{I} - \underline{e}_3 \otimes \underline{e}_3) + \sin \theta \underline{R}_3$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \cos \theta \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \sin \theta \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\underline{Q}(\underline{e}_3, \theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotate $\underline{e}_1$ by $90^\circ (\frac{\pi}{2})$ counter clockwise

$$\cos\left(\frac{\pi}{2}\right) = 0 \quad \sin\left(\frac{\pi}{2}\right) = 1$$

$$\underline{\underline{Q}}(\underline{\underline{e}}_3, \frac{\pi}{2}) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\underline{\underline{Q}}(\underline{\underline{e}}_3, \frac{\pi}{2}) \underline{\underline{e}}_1 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \underline{\underline{e}}_2 \quad \checkmark$$

Infinitesimal Rotations

$$\lim_{\theta \rightarrow 0} \underline{\underline{Q}}(\underline{\underline{r}}, \theta) = (\underline{\underline{r}} \otimes \underline{\underline{r}}) + \cancel{\cos \theta}^1 (\underline{\underline{I}} - \underline{\underline{r}} \otimes \underline{\underline{r}}) + \cancel{\sin \theta}^\theta \underline{\underline{R}} \\ = \underline{\underline{I}} + \theta \underline{\underline{R}}$$

$\Rightarrow$ Axial tensor $\underline{\underline{R}}$ give infinitesimal rotation

$$\underline{\underline{v}} = (\underline{\underline{I}} + \theta \underline{\underline{R}}) \underline{\underline{u}}$$

$$\underline{\underline{v}} = \underline{\underline{u}} + \theta (\underline{\underline{r}} \times \underline{\underline{u}})$$

$\Rightarrow$ cross product gives infinitesimal rotation

Determine θ and $\underline{r}$ from $\underline{Q}$:

1) Find rotation angle

$$\text{tr}(\underline{Q}) = Q_{ii} = r_i r_i + \cos \theta (\delta_{ii} - r_i r_i) + \sin \theta \epsilon_{ikj} r_k$$

$$r_i r_i = \underline{r} \cdot \underline{r} = 1 \quad \text{unit vector}$$

$$\delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} = 3$$

$$\epsilon_{iki} = 0$$

$$\Rightarrow \text{tr}(\underline{Q}) = 1 + 2 \cos \theta \Rightarrow \cos \theta = \frac{\text{tr}(\underline{Q}) - 1}{2}$$

Example: $\underline{Q}(\underline{e}_3, \frac{\pi}{2}) \quad \text{tr}(\underline{Q}) = 1$

$$\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

2) Axis of rotation $\underline{r}$:

$$\underline{R} = \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix} \quad \text{is axial tensor}$$

we are given $\underline{Q}$ not $\underline{R}$

$\Rightarrow$ Extract $\underline{R}$ from $\underline{Q}$

$$\underline{\underline{Q}} = \text{sym}(\underline{\underline{Q}}) + \text{skew}(\underline{\underline{Q}})$$

$$\text{sym}(\underline{\underline{Q}}) = \frac{1}{2}(\underline{\underline{Q}} + \underline{\underline{Q}}^T) = \underline{\underline{r}} \otimes \underline{\underline{r}} + \cos \theta (\underline{\underline{I}} - \underline{\underline{r}} \otimes \underline{\underline{r}})$$

$$\text{skew}(\underline{\underline{Q}}) = \frac{1}{2}(\underline{\underline{Q}} - \underline{\underline{Q}}^T) = \sin \theta \underline{\underline{R}} = \sin \theta \epsilon_{ikj} r_k \underline{\underline{e}}_i \otimes \underline{\underline{e}}_j$$

$$\frac{1}{2}(\underline{\underline{Q}} - \underline{\underline{Q}}^T) = \frac{1}{2}(Q_{ij} - Q_{ji}) \underline{\underline{e}}_i \otimes \underline{\underline{e}}_j$$

equate two expressions for components

$$\underbrace{\frac{1}{2}(Q_{ij} - Q_{ji})}_{\text{know}} = \sin \theta \epsilon_{ikj} \underbrace{r_k}_{\text{want}}$$

remove ϵ_{ikj} using $\epsilon \delta$ identities

$$\begin{aligned} \epsilon_{ilj} \frac{1}{2}(Q_{ij} - Q_{ji}) &= \sin \theta \epsilon_{ilj} \epsilon_{ikj} r_k \\ &= \sin \theta \epsilon_{lij} \epsilon_{kij} r_k \\ &= \sin \theta 2 r_l \end{aligned}$$

$$\boxed{\epsilon_{lij} \epsilon_{kij} = 2 \delta_{lk}}$$

$$\Rightarrow \quad \underline{\underline{r}}_l = \frac{\epsilon_{ilj}(Q_{ij} - Q_{ji})}{4 \sin \theta} \quad \underline{\underline{r}} = \frac{1}{2 \sin \theta} \begin{bmatrix} Q_{32} - Q_{23} \\ Q_{13} - Q_{31} \\ Q_{21} - Q_{12} \end{bmatrix}$$

Example: $Q(\underline{e}_3, \frac{\pi}{2}) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\Gamma = \frac{1}{2 \cancel{\sin(\frac{\pi}{2})}} \begin{bmatrix} 0 & -0 \\ 0 & -0 \\ 1 & +1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \underline{e}_3$